

4. Data Preprocessing

Data preprocessing was performed to create new useful columns for better analysis and dashboard creation.

Purpose of Data Preprocessing

The main goal was to:

- Organize data properly
 - Create useful categories
 - Make analysis easier
 - Prepare data for PivotTables and charts
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What Was Done

- Inserted new columns
 - Created Age Group column
 - Created Month column
 - Applied formulas
 - Converted formulas into values
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Inserting New Column

A new column was inserted between existing columns.

Steps Performed

1. Selected the column.
2. Pressed:

Ctrl + Shift + +

3. A new blank column was created.

Purpose

Used to create new calculated columns.

Creating Age Group Column

A new column named:

Age Group

was created.

Formula Used

=IF(E2>=50,"Senior",IF(E2>=30,"Adult","Teenager"))

Logic of Formula

- Age greater than or equal to 50 → Senior
- Age greater than or equal to 30 → Adult
- Otherwise → Teenager

Steps Performed

1. Entered formula in the first cell.
2. Pressed Enter.
3. Moved cursor to bottom-right corner of cell.
4. Double-clicked to autofill formula for all rows.

Purpose

Automatically categorized customers into different age groups.

Benefit

Helped perform:

- Age analysis
- Age vs Gender analysis
- Customer segmentation

Creating Month Column

Another new column named:

Month

was created.

Formula Used

=TEXT(G2,"mmm")

Result

Converted dates into month names:

- Jan
- Feb
- Mar

Steps Performed

1. Entered formula in first cell.
2. Pressed Enter.
3. Used autofill to apply formula to all rows.

Purpose

Extracted month names from dates for monthly sales analysis.

Benefit

Helped create:

- Monthly sales charts
- Orders vs Sales analysis
- Monthly trend analysis

Converting Formula into Values

After formulas were applied:

1. Selected entire column.
2. Copied the column.
3. Used:

Paste Special → Values

Purpose

Removed formulas and kept only final values.

Benefit

Improved dataset stability and prevented formula issues during analysis.